-: Experiment 2:-

Aim:-

Convert the ER diagram into a relational scheme. Write complete MySQL Create Table statements with primary keys, foreign key, NOT NULL, UNIQUE, ON DELETE CASCADE and SET NULL Constraints. Insert sample data and demonstrate referential integrity violations.

1. Relational Schema

Table	Primary key	Important Foreign keys
CUSTOMER	Customer ID	—
ADDRESS	Address ID	customer ID → CUSTOMER
SELLER	Seller Id	—
CATEGORY	Category ID	Seller ID → SELLER
PRODUCT	Product ID	Category ID – CATEGORY
ORDERS	Order ID	Customer ID – CUSTOMER
ORDER_ITEM	Order Item ID	Order ID → ORDERS Product ID → PRODUCT
PAYMENT	Payment ID	Order ID – ORDERS
DELIVERY	Delivery ID	Order ID – ORDERS Address ID – ADDRESS

```
CREATE TABLE CATEGORY(
Catagory ID INT PRIMARY KEY,
Category Name VARCHAR(50) NOT NULL,
Seller ID INT,
FOREIGN KEY (sellerID) REFERENCES
  SELLER (sellerID)
   ON DELETE SET NULL
 );

CREATE TABLE PRODUCT(
Product ID INT PRIMARY KEY,
Product Name VARCHAR(50) NOT NULL,
Price DECIMAL(10.2) NOT NULL,
Stock Quantity INT NOT NULL,
Category ID INT,
FOREIGN KEY (Category ID)
REFERENCES CATEGORY (CategoryID)
  ON DELETE SET NULL
);

CREATE TABLE ORDERS(
 Order ID INT PRIMARY KEY,
 Order DATE NOT NULL,
 Total Amount DECIMAL (10.2) NOT NULL,
 Customer ID INT NOT NULL,
 FOREIGN KEY (Customer ID)
 REFERENCES CUSTOMER (customer ID)
    ON DELETE CASCADE
);
```

2. SQL Queries

```
CREATE DATABASE ECOMMERCE
 USE ECOMMERCE

CREATE TABLE CUSTOMER
 Customer ID  INT PRIMARY KEY
 Name  VARCHAR(50) NOT NULL
 Password  VARCHAR(100) NOT NULL,
 Phone  VARCHAR (15) UNIQUE NOT
  NULL
 Email  VARCHAR(100) UNIQUE NOT
   NULL
 );
 CREATE TABLE ADDRESSC
 Address ID INT PRIMARY KEY
 PIN  VARCHAR (10)  NOT NULL
 State VARCHAR (30) NOT NULL
 Billing shipping  VARCHAR (30),
 Customer ID INT NOT NULL,
  FOREIGN KEY  customerID
 REFERENCES CUSTOMER (CustomerID)
     ON DELETE CASCADE
 );
 CREATE TABLE SELLERC
   Seller ID INT PRIMARYKEY
   Seller Name VARCHAR (50) NOT NULL
  Email  VARCHAR(100)  UNIQUE NOT
    NULL,
  Phone  VARCHAR (15)  UNIQUE NOT NULL
  );
```

```
CREATE TABLE ORDER_ITEM(
 Order ItemID INT PRIMARY KEY,
  Quanity  INT NOT NULL,
  Price DECIMAL (10,2) NOT NULL,
  OrderID  INT NOT NULL,
  Product ID INT NOT NULL,
  FOREIGN KEY (orderID) REFERENCES
   ORDERS (orderID)
  ON DELETE CASEADE,
  FOREIGN KEY(ProductID) REFERENCES
   PRODUCT (ProductID)
   ON DELETE CASCADE
  );

  CREATE TABLE PAYMENT(
  PaymentID  INT PRIMARY KEY,
  Amount  DECIMAL (10.2) NOT NULL,
 Payment Date DATE NOT NULL
 Payment Method  VARCHAR (30) NOT
   NULL,
   Order ID  INT UNIQUE NOT NULL
   FOREIGN KEY (orderID) REFERENCES
   ORDERS (order ID)
    ON DELETE CASECADE
  );
CREATE  TABLE DELIVERY (
   DeliveryID INT PRIMARY KEY,
   Delivery Date  DATE,
  Status  VARCHAR (30) NOT NULL
  Order ID INT UNIQUE NOT NULL
   Address INT,
   FOREIGN KEY (orderID) REFERENCES
   ORDERS (orderID)
     ON DELETE CASECADE,
   FOREIGN KEY (AddressID) REFERENCES
   ADDRESS (AddressID)
  );ON DELETE SET  NULL
```

3. Insert Sample Data

```
Insert into customer values
(1, 'Divija', 'Pass123', '9876543210', 'divija@gmail.com');

Insert into Address values
(101, '160017', 'Punjab', 'Home', 1);

Insert into Seller values
(201, 'Techworld', techworld@gmail.com, '9123456789');

Insert into category values
(301, 'Electronics', 201);

Insert into product values
(401, 'Laptop', 75000, 10, 301);

Insert into orders values
(501, '2026-08-14', 75000, 1);

Insert into order_item
   values
(601, 1, 75000, 501, 401);
```

Insert into Payment values
(701, 75000, '2026-08-14', UPI', 501);

Insert into delivery values
(801, NULL, 'Processing', 501, 101);

4. Referential Integrity Violation

Insert into orders values
(502, '2026-08-14', 50000, 999);

Output:

Error 1452 (23000);
Cannot add or update a child row; a foreign key constraint fails

5. On delete set Null

Select Product ID, Product name,
Category ID
From product;
Output

Product ID	Product Name	Category ID
401	Laptop	301

Delete from Category
where Category ID = 301;

Select product ID, Product Name, Category ID
From product

Output

Product ID	Product Name	Category ID
401	Laptop	NULL

6. ON Delete Cascade

Select * From orders

Output:

Order ID	Order Date	Total Amount	Customer ID
501	2026-8-14	75000.00	1

Delete from Customer
where customer ID = 1;

Select * from orders;
Output:
Empty set

Output:

Show Tables;

Output:

Address
Category
Customer
Delivery
Orders
Order_Item
Payment
Product
Seller

Select * From Customer

Output:

CustomerID	Name	Password	Phone	Email
1.	Divija	Pass123	9876543210	divija@gmail.com

Select from Product;

Output:

Product ID	Product Name	Price	Stock Quantity	Category ID
401	Laptop	75000.00	10	301

Result:

The relational schema was successfully implemented and the required constraints were demonstrated.

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